Degree-two zero-B point extraction

Component result; independent review pending.

The retained source branch is `double22_weighted_variant`, $B = 0$, $s \neq 0$. Six Lean declarations compiled; fixed controls and separate axiom readback passed. The declarations report only `propext`, `Classical.choice` and `Quot.sound`.

Statements and assumptions

From original normalized Mumford validity, nonzero input weights, $s \neq 0$, characteristic different from two and the literal $B = 0$ condition, the extracted point has nonzero weights and lies on the curve. Extraction requires no smoothness assumption. A separate theorem adds global coprimality of the quintic and its derivative, derives a nonzero point ordinate, and composes the reviewed weighted doubling validity result.

Fixed controls

The characteristic-five positive uses $f = X^5+X^2-X$, $u = X^2-X$, $v = X$, weighted input $(1, 0, 4, 0, 2, 3)$. Its extracted point is $(4, 4, 3, 3)$, normalizing to $(1, 1)$. The explicit coprimality certificate is $4f+3Hf' = 1$, with $H = X^4 + 3X^3 + 4X^2 + 3X + 3$ over $\mathbb{F}_5$.

For the singular rational input $f = X^5$, $u = X^2$, $v = X$, the extracted ordinate and following doubling weight are zero. In characteristic two, $f = X^5 + X^2 + X + 1$, $u = X^2 + X + 1$, $v = X$ satisfy original validity and the branch conditions, but the extracted point is invalid. A zero-s input demonstrates the nonzero-weight boundary.

Evidence and remaining review

Main compilation: attempt4; controls: controls5; readback: readback1. Earlier attempts and control-script failures remain retained. Two tactic-style warnings remain in the passing controls. The companion JSON binds source hashes, dependency inventory and every receipt. No scientific payload or cost measurement was run. No UInt64, compiler, class-sum or counter correspondence is asserted by this field-level packet. Independent source-fidelity review is pending; the module is not integrated into the canonical project.